

Polynomial Regression

Wednesday, June 24, 2026 10:04 PM

Years
1
2
3
4

Salary
50K
55K
65K
75K

$y = ax^2 + bx + c$
Years

Target $\rightarrow$ numerical

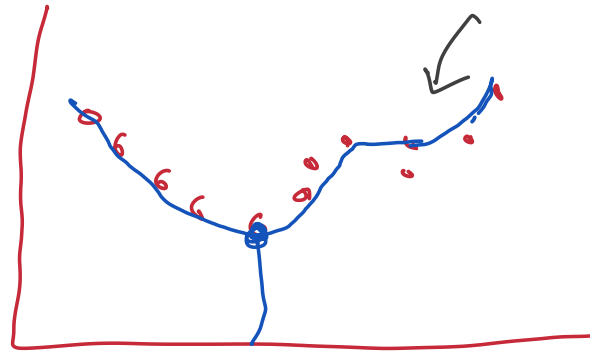
non linear data

PR

Better fit for curved data.

$y = ax^2 + bx + c$

Dataset shape goes as
 $\rightarrow$ scatter plot method



$$y = a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

$n = \text{degree}$

Degree	Train R ²	Validation R ²
1	0.72	0.70
2	0.88	0.86

3	0.91	0.84
4	0.95	0.78
5	0.98	0.62

এখানে **Degree 2** best choice হতে পারে।

কারণ degree 3+ training performance বাড়াতে validation performance কমে।

21. Polynomial Regression বনাম Linear Regression

বিষয়	Linear Regression	Polynomial Regression
Relationship	Linear	Nonlinear/curved
Equation	$b_0 + b_1x$	$b_0 + b_1x + b_2x^2 \dots$
Complexity	Low	Higher
Overfitting risk	Low	Higher
Interpretability	সহজ	তুলনামূলক কঠিন
Training	Fast	Degree বাড়লে complex
Feature engineering	কম	Polynomial features দরকার

